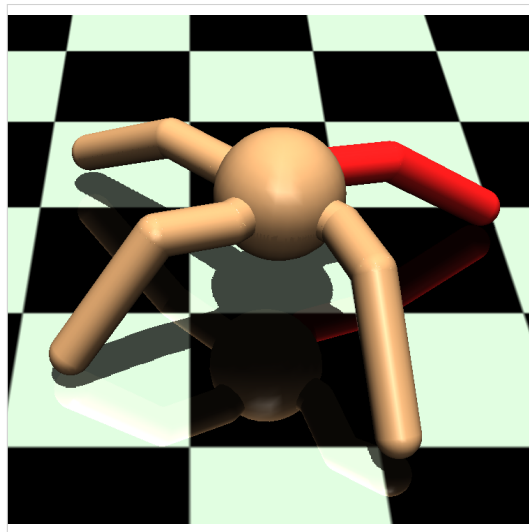


Damage-Robust Locomotion of a Quadraped Robot Using Reinforcement Learning

AE4350: Bio-inspired Intelligence and Learning for Aerospace Applications

Tommaso Calzolari

Student number: 6430600



Research question:

How does training under randomized single leg actuator degradation affect the robustness and healthy performance of a PPO locomotion controller?

GitHub repository:

github.com/tommasocalzolari/damage-robust-ant-rl

Video demonstrations: [Open Google Drive folder](#)

Contents

1	Introduction	1
1.1	Motivation and related work	1
1.2	Research question and approach	1
2	Problem Formulation	1
2.1	The Ant-v5 locomotion task	1
2.2	Actuator degradation model	2
2.3	Training distributions	2
3	Method and Experimental Design	2
3.1	Proximal Policy Optimization	2
3.2	Why the original PPO setup failed	2
3.3	Final PPO configuration	3
3.4	Training and evaluation protocol	4
3.5	Post-hoc fine-tuning sensitivity	4
4	Results and Discussion	4
4.1	Learning behaviour	4
4.2	Nominal performance and damage robustness	5
4.3	Leg-specific response and command use	5
4.4	Sensitivity and selected policy	6
4.5	Limitations	6
4.6	Conclusion	6
	Bibliography	8

Abstract

This study investigates whether exposing a locomotion policy to randomized single-leg actuator degradation during training improves its performance after damage. Proximal Policy Optimization (PPO) policies were trained in Gymnasium’s MuJoCo Ant-v5 environment under nominal and randomized-damage conditions. An initial PPO design failed to learn sustained locomotion because policy updates became too large. Observation and reward normalization, a decaying learning rate, a KL-divergence limit and a larger survival reward stabilized learning. Three fresh policies per condition were then trained for five million environment steps and evaluated under healthy operation, half strength and complete command loss in each leg. Robust training increased mean horizontal distance under both damage severities, but variation between the three training seeds was large. A post-hoc sensitivity study improved the best robust policy further, although this one-seed result is kept separate from the main comparison.

Introduction

Legged locomotion requires coordinated actuation under nonlinear, intermittent ground contact. Policies trained only on an intact robot may exploit gaits that fail when actuator capability changes. This report tests whether exposure to actuator damage during training improves post-damage locomotion.

1.1. Motivation and related work

Reinforcement learning (RL) can learn continuous locomotion directly through interaction rather than through a prescribed gait. Policy-gradient methods have been applied successfully to quadrupedal locomotion [3, 5]. Generalized Advantage Estimation (GAE) reduces the variance of policy-gradient estimates [8], while PPO limits the incentive for excessively large policy updates through a clipped objective [9].

Robustness can be encouraged by training over a distribution of dynamics. Dynamics randomization has improved transfer across changing physical parameters [6], while more advanced methods infer hidden properties online [4] or search for a new behaviour after damage [1]. The present experiment is intentionally simpler: there is no fault identifier, online adaptation or post-damage learning. A single frozen PPO policy must respond to damage using only its normal state observation.

1.2. Research question and approach

The central research question is

How does training under randomized single-leg actuator degradation affect the robustness and healthy performance of a PPO locomotion controller?

Nominal policies were trained on an intact Ant; robust policies encountered episodes that were healthy or had a persistent reduction in both actuators of one randomly selected leg. Three policies per condition were evaluated with common initial-state seeds under healthy operation, 50% strength and complete command loss. Horizontal distance and early termination were the primary outcomes, supported by returns and command magnitudes.

Problem Formulation

2.1. The Ant-v5 locomotion task

The experiments use Gymnasium’s MuJoCo Ant-v5 environment [2, 12]. The Ant is a three-dimensional quadruped with four two-link legs and eight actuated hinge joints. Its 105-dimensional observation comprises 13 generalized positions, 14 generalized velocities and 78 clipped external contact-force components. Global torso x and y coordinates are excluded from the policy observation but retained for evaluation. The action is

$$a_t \in [-1, 1]^8. \quad (2.1)$$

Each action is held for five 0.01 s MuJoCo frames, giving $\Delta t = 0.05$ s. Episodes last at most 1000 steps and end early if the state becomes non-finite or torso height leaves $[0.2, 1.0]$ m.

The final reward retained the standard forward, control and contact terms, but increased the healthy reward from 1 to 3:

$$r_t = 3 + \frac{x_{t+1} - x_t}{\Delta t} - 0.5 \|\ddot{a}_t\|_2^2 - 5 \times 10^{-4} \|F_t^{\text{clip}}\|_2^2. \quad (2.2)$$

The larger survival term makes falling costly, while horizontal distance is reported separately so that standing still is not mistaken for successful locomotion.

2.2. Actuator degradation model

The action wrapper scales both commands of one leg before they reach MuJoCo. For damaged leg ℓ ,

$$\tilde{a}_t = D_{\ell, \alpha} a_t, \quad D_{\ell, \alpha} = \text{diag}(d_0, \dots, d_7), \quad d_i = \begin{cases} \alpha, & i \in \mathcal{I}_\ell, \\ 1, & i \notin \mathcal{I}_\ell. \end{cases} \quad (2.3)$$

Here $\alpha = 1$ is healthy, $\alpha = 0.5$ gives half strength and $\alpha = 0$ removes both commands. The damage is sampled at reset and remains fixed for the episode. The policy receives no explicit damage label. Table 2.1 gives the actuator mapping obtained from the MuJoCo model.

Table 2.1: Leg and actuator mapping used by the damage wrapper.

Leg	Action indices	Target joints
front_left	2, 3	hip_1, ankle_1
front_right	4, 5	hip_2, ankle_2
back_left	6, 7	hip_3, ankle_3
back_right	0, 1	hip_4, ankle_4

The wrapper records both a_t and $\tilde{a}_t$. Their mean absolute values are command-magnitude proxies, not physical energy measurements.

2.3. Training distributions

Nominal training leaves all commands unchanged. During robust training, an episode is healthy with probability 0.25. Otherwise one of the four legs is selected uniformly and

$$\alpha \sim \mathcal{U}(0.25, 1). \quad (2.4)$$

Complete failure, $\alpha = 0$, lies outside this distribution and therefore tests generalization beyond the damage observed during training.

3

Method and Experimental Design

3.1. Proximal Policy Optimization

PPO is an on-policy actor–critic method [9]. The actor defines $\pi_\theta(a_t | s_t)$, while the critic estimates $V_\phi(s_t)$. Advantages are computed with GAE,

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \quad \hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma \lambda)^l \delta_{t+l}, \quad (3.1)$$

and the actor maximizes the clipped surrogate objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)], \quad (3.2)$$

where $r_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$. The experiments used the PPO implementation in Stable-Baselines3 [7, 10]. Training diagnostics, including approximate KL divergence, clipping fraction, return and episode length, were inspected in TensorBoard through its Stable-Baselines3 integration [11].

3.2. Why the original PPO setup failed

The first comparison used unnormalized Ant observations and rewards, a constant 3×10^{-4} learning rate and no KL-divergence limit. One-million-step policies usually lunged and fell; extending selected seeds to five million steps made this behaviour worse. In healthy evaluation, 96.7% of nominal and 83.3% of robust episodes ended early. At roughly one million steps, mean approximate KL was 0.608 ± 0.582 and clipping

fraction was 0.809 ± 0.115 , showing that updates were no longer small “proximal” changes. Mean episode duration was only 88.7 ± 66.0 steps.

Four changes were made together: observations and training rewards were normalized and clipped to ± 10 , the learning rate decayed linearly from 3×10^{-4} to 10^{-4} , a target KL of 0.02 limited large updates, and the healthy reward was raised to 3. At one million steps, mean KL fell to 0.0104 ± 0.0010 , clipping fraction to 0.137 ± 0.015 , and episode duration rose to 423.9 ± 56.6 steps. A reward-5 pilot survived but barely moved, so reward 3 was retained. Because several elements changed together, this diagnostic does not identify one causal fix.

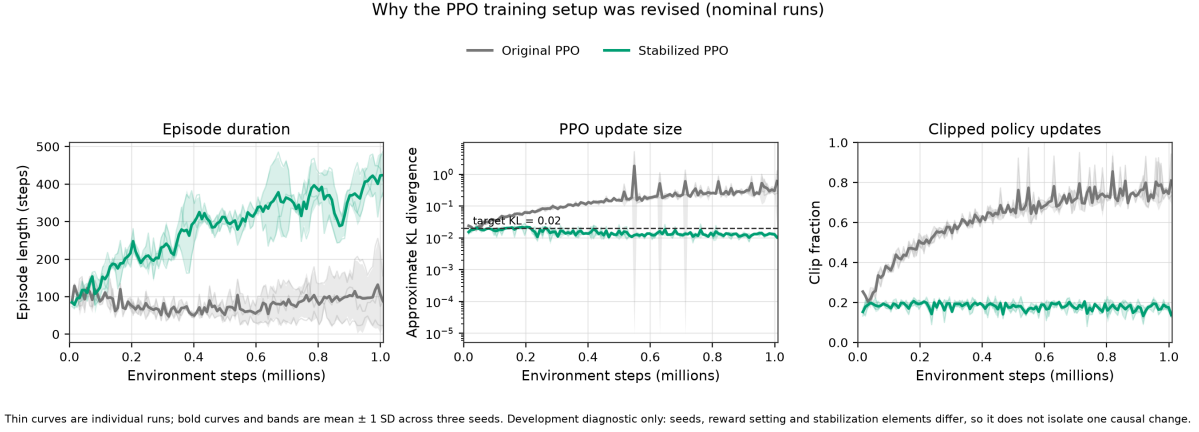


Figure 3.1: Development diagnostic comparing the failed and stabilized PPO designs over their first million steps. Episode-duration curves use the Stable-Baselines3 100-episode statistics window, or all available episodes before the buffer fills. KL-divergence and clip-fraction points are unsmoothed update aggregates. Curves show three runs per design; several settings and seeds differ, so this is not a one-factor experiment.

3.3. Final PPO configuration

Both final conditions used the same network and PPO settings; only the damage distribution differed. The policy contained separate two-layer actor and critic networks with 256 Tanh units per layer. Table 3.1 summarizes the configuration.

Table 3.1: PPO configuration used in the final comparison.

Parameter	Value	Parameter	Value
Policy	MlpPolicy	Parallel environments	4
Actor / critic	256, 256 each	Steps per environment	2048
Activation	Tanh	Batch size	64
Learning rate	Linear, $3 \times 10^{-4} \rightarrow 10^{-4}$	Epochs	10
Discount γ	0.99	GAE λ	0.95
Clip range ϵ	0.2	Target KL	0.02
Entropy coefficient	0	Gradient norm	0.5
Healthy reward	3.0	Requested steps	5×10^6

With four environments, each rollout contains 8192 transitions; completed rollouts produced 5,005,312 transitions per requested five-million-step run. With the remaining training fraction p decreasing from 1 to 0, the exact learning-rate schedule was $\eta(p) = 10^{-4} + 2 \times 10^{-4}p$.

Stable-Baselines3 VecNormalize used `norm_obs=True`, `norm_reward=True`, `clip_obs=clip_reward=10`, $\gamma = 0.99$ and $\epsilon = 10^{-8}$. It standardized observations using running means and variances and scaled training rewards by the running standard deviation of discounted returns. At evaluation, observations were normalized using frozen statistics (`training=False`), while reward normalization was disabled (`norm_reward=False`).

The runs used Python 3.12.3, Gymnasium 1.3.0, MuJoCo 3.11.0 and Stable-Baselines3 2.9.0.

3.4. Training and evaluation protocol

Fresh nominal and robust policies were trained with seeds 5, 6 and 7. The six policies were evaluated deterministically using common episode seeds 300–309 under nine conditions: healthy operation, every leg at $\alpha = 0.5$, and every leg at $\alpha = 0$. This produced 540 episodes.

Horizontal distance was selected as the main locomotion measure because a stable gait in any ground-plane direction was acceptable:

$$d_{xy} = \sqrt{(x_T - x_0)^2 + (y_T - y_0)^2}. \quad (3.3)$$

An episode was marked as an early termination when the Ant became unhealthy before 1000 steps. Results were averaged over episodes within each trained policy. Damaged legs were then averaged within each severity, followed by the mean and sample standard deviation across the three training seeds. Evaluation episodes were not treated as independent training repetitions.

3.5. Post-hoc fine-tuning sensitivity

An exploratory 2×2 study continued copies of the best robust policy (seed 6) for 1,007,616 transitions under the robust damage distribution, using constant learning rates and clip ranges

$$\eta \in \{10^{-4}, 3 \times 10^{-4}\}, \quad \epsilon \in \{0.1, 0.2\}. \quad (3.4)$$

On seeds 400–409, configurations were ranked by mean signed forward displacement $x_T - x_0$ over 40 half-strength episodes, comprising ten seeds for each of four damaged legs. The tie breakers were termination rate, forward speed and healthy signed displacement; this selection grid did not use d_{xy} . The selected policy was then compared with its source on unused seeds 500–509 using horizontal displacement d_{xy} . This one-seed, post-hoc study refines a demonstration policy and is separate from the six-policy comparison.

4

Results and Discussion

4.1. Learning behaviour

Returns increased over five million steps in both conditions (Figure 4.1). Robust return includes randomized damage and is therefore not directly comparable with nominal return.



Thin lines: individual seeds. Bold line and band: mean $\pm$ 1 SD across three training seeds. Each seed curve is SB3's trailing mean over 100 completed episodes; robust returns include randomized damage.

Figure 4.1: Episode return during the six final runs. Each thin seed curve is the Stable-Baselines3 mean over the 100 most recently completed episodes, or all episodes before the buffer filled. Bold lines and bands show mean $\pm$ one sample standard deviation across the three seed curves.

4.2. Nominal performance and damage robustness

Table 4.1 summarizes the primary results. Robust training left healthy mean distance almost unchanged and increased mean distance from 24.55 to 30.42 m at half strength and from 8.07 to 10.06 m under complete failure, gains of 23.9% and 24.6%. All damaged episodes reached the time limit. Healthy robust policies had a 6.7% early-termination rate, compared with 0% for nominal policies.

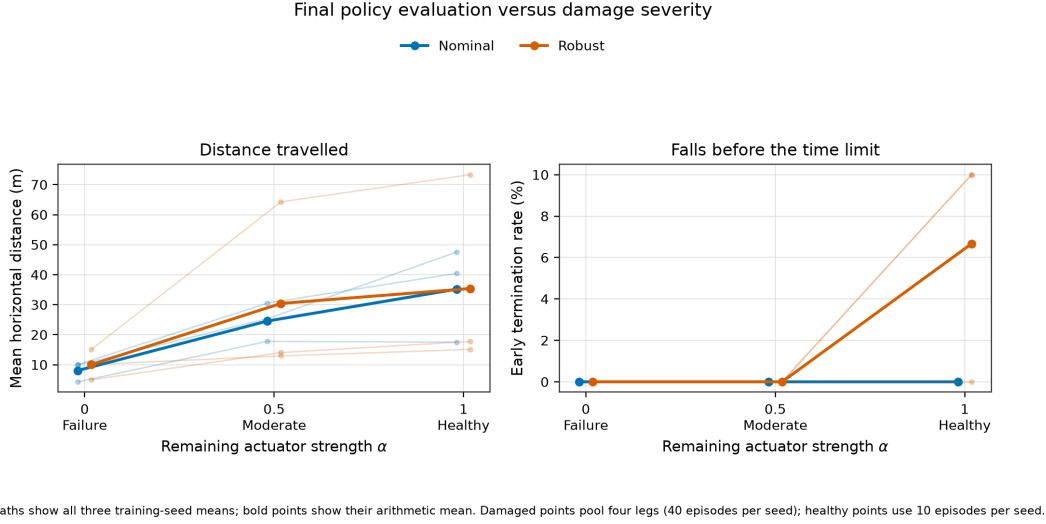


Figure 4.2: Horizontal distance and early termination across damage severity. Faint paths show individual training seeds and bold paths their mean.

Table 4.1: Final deterministic evaluation. Distances are mean $\pm$ sample standard deviation across three training-seed means. Termination rates aggregate 30 healthy or 120 damaged episodes per training condition.

(lr)2-3(lr)4-5 Remaining strength	Horizontal distance (m)		Early termination	
	Nominal	Robust	Nominal	Robust
Healthy	35.17 $\pm$ 15.69	35.40 $\pm$ 32.87	0.0%	6.7%
$\alpha = 0.5$	24.55 $\pm$ 6.42	30.42 $\pm$ 29.32	0.0%	0.0%
$\alpha = 0$	8.07 $\pm$ 3.26	10.06 $\pm$ 5.08	0.0%	0.0%

The uncertainty is important. Robust seed 6 travelled 73.32 m when healthy and 64.28 m at half strength, compared with robust three-seed means of 35.40 and 30.42 m. Nominal seed 6, the best nominal policy, travelled 47.53 and 25.34 m. The best robust behaviour is clear, but a reliable improvement across random initializations is not established.

4.3. Leg-specific response and command use

Damage did not affect every leg equally. At half strength, robust mean distance was higher for every leg, with the largest difference for the front-right leg (33.11 versus 16.51 m). Under complete failure, robust training helped strongly for the front-right leg but not for every other leg. This asymmetry and the spread between seeds require a cautious overall conclusion.

For front-left half-strength damage, the robust policy requested a larger raw command from the damaged leg (0.295 versus 0.245 for nominal) and used more similar command magnitudes across the three intact legs (Figure 4.3). Only the selected leg is scaled, so its applied command is exactly half its raw command. Together with the distance result, this is consistent with compensation through a stronger request to the weakened leg and more even command use across the intact legs. Command averages do not measure forces or establish a specific gait mechanism.

The selected demonstration video provides a qualitative check of this interpretation. In the refined robust seed-6 clip (complete_damage.mp4) with evaluation seed 502, front-right damage and $\alpha = 0$, the disabled limb mostly trails and makes passive ground contact, while the three functioning legs alternate larger planting

4 Results and Discussion

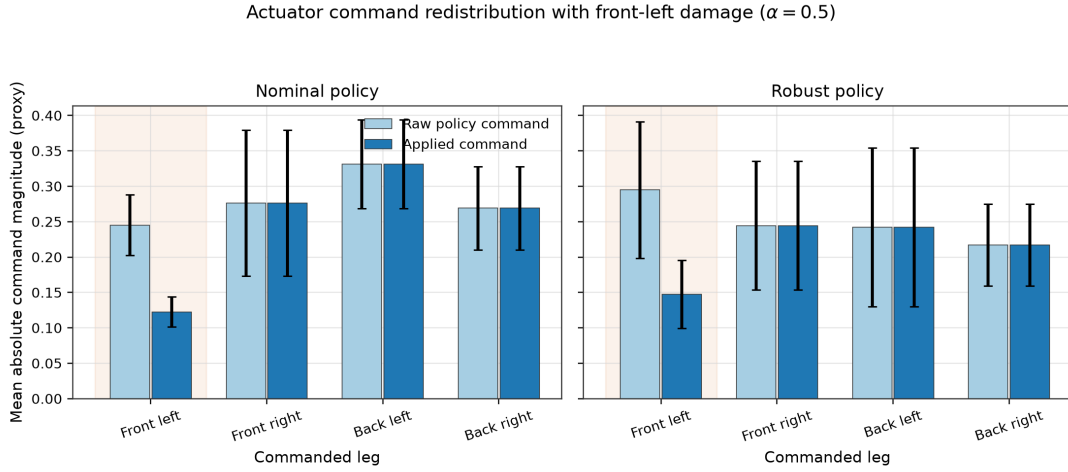


Figure 4.3: Mean raw and applied commands during front-left half-strength damage. Bars show mean $\pm$ one sample standard deviation across the three training seeds.

and recovery sweeps. The gait is visibly asymmetric rather than a recovered four-leg pattern. The torso remains elevated but rocks and yaws during each cycle. These observations are consistent with three-limb compensation, but a selected video cannot establish how often the same strategy occurs.

The corresponding clips are available in the [video demonstration archive](#).

4.4. Sensitivity and selected policy

The sensitivity grid in Figure 4.4 selected $\eta = 10^{-4}$ and $\epsilon = 0.1$. Its validation mean signed forward displacement, pooled over the four half-strength legs, was 65.10 m with 5% early termination; the weakest combination, 3×10^{-4} and 0.2, achieved 43.01 m. For this selected seed, settings that permitted smaller updates were associated with better validation performance, consistent with the earlier stability diagnosis.

On held-out seeds, the refined policy travelled 67.96 m when healthy and 73.68 m averaged over the four half-strength legs. The original robust seed-6 policy achieved 60.16 and 57.15 m on the same seeds. Under complete failure, the pooled means were 22.15 and 17.78 m, but improvements were not uniform for every leg (Figure 4.4, lower panel). The refinement should therefore be interpreted as a selected one-seed demonstration, not additional evidence for the main comparison.

4.5. Limitations

Evidence is limited by three training seeds and the dominance of robust seed 6. Damage only scaled commands in one fixed leg per episode; sensor faults, changing damage and hardware effects were absent. Horizontal distance accepts motion in any direction and cannot distinguish walking from sliding, so videos provided a qualitative check. The PPO correction changed several elements together, and the sensitivity study continued one selected seed. The results therefore establish a controlled simulation outcome, not general fault-tolerant locomotion.

4.6. Conclusion

After PPO stabilization, robust training preserved average healthy distance and increased mean horizontal distance by 23.9% at half strength and 24.6% under complete command loss. The command pattern of the best robust policy was consistent with compensation, but did not identify a gait mechanism. Large seed-to-seed variation means that the evidence supports a possible robustness benefit rather than a reliable improvement across random initializations. More training seeds and a factorial study of the PPO corrections would provide a stronger test.

4 Results and Discussion

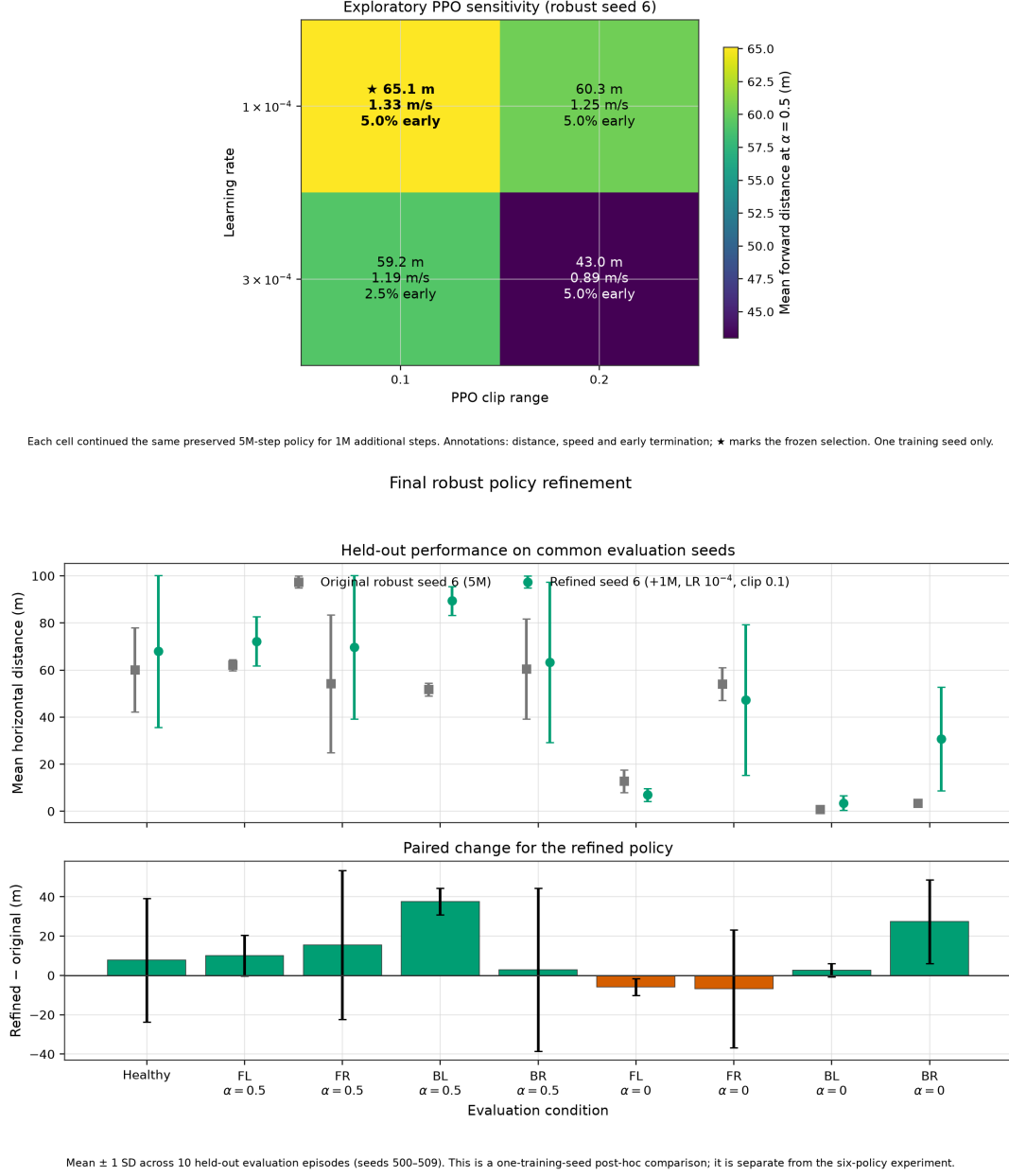


Figure 4.4: Exploratory sensitivity and held-out evaluation for robust seed 6. The upper grid reports mean signed forward displacement $x_T - x_0$ for four one-million-step continuations; the star marks the selected configuration. The lower plot compares that refinement with its preserved five-million-step source on common held-out seeds using horizontal displacement d_{xy} .

Bibliography

- [1] Antoine Cully et al. 'Robots that Can Adapt Like Animals'. In: *Nature* 521 (2015), pp. 503–507. doi: [10.1038/nature14422](https://doi.org/10.1038/nature14422).
- [2] Farama Foundation. *Ant — Gymnasium Documentation*. <https://gymnasium.farama.org/environments/mujoco/ant/>. Accessed 14 August 2026. 2026.
- [3] Jemin Hwangbo et al. 'Learning Agile and Dynamic Motor Skills for Legged Robots'. In: *Science Robotics* 4.26 (2019), eaau5872. doi: [10.1126/scirobotics.aau5872](https://doi.org/10.1126/scirobotics.aau5872).
- [4] Ashish Kumar et al. 'RMA: Rapid Motor Adaptation for Legged Robots'. In: *Robotics: Science and Systems XVII*. 2021. doi: [10.15607/RSS.2021.XVII.011](https://doi.org/10.15607/RSS.2021.XVII.011).
- [5] Joonho Lee et al. 'Learning Quadrupedal Locomotion over Challenging Terrain'. In: *Science Robotics* 5.47 (2020), eabc5986. doi: [10.1126/scirobotics.abc5986](https://doi.org/10.1126/scirobotics.abc5986).
- [6] Xue Bin Peng et al. 'Sim-to-Real Transfer of Robotic Control with Dynamics Randomization'. In: *2018 IEEE International Conference on Robotics and Automation*. 2018, pp. 3803–3810. doi: [10.1109/ICRA.2018.8460528](https://doi.org/10.1109/ICRA.2018.8460528).
- [7] Antonin Raffin et al. 'Stable-baselines3: reliable reinforcement learning implementations'. In: *J. Mach. Learn. Res.* 22.1 (Jan. 2021). ISSN: 1532-4435.
- [8] John Schulman et al. 'High-Dimensional Continuous Control Using Generalized Advantage Estimation'. In: *International Conference on Learning Representations*. 2016.
- [9] John Schulman et al. 'Proximal Policy Optimization Algorithms'. In: *arXiv preprint arXiv:1707.06347* (2017).
- [10] Stable-Baselines3 Contributors. *Stable-Baselines3 Documentation: Reliable Reinforcement Learning Implementations*. Software documentation. URL: <https://stable-baselines3.readthedocs.io/en/master/> (visited on 28/08/2026).
- [11] Stable-Baselines3 Contributors. *TensorBoard Integration*. Stable-Baselines3 documentation. URL: <https://stable-baselines3.readthedocs.io/en/master/guide/tensorboard.html> (visited on 28/08/2026).
- [12] Emanuel Todorov, Tom Erez and Yuval Tassa. 'MuJoCo: A physics engine for model-based control.' In: *IROS*. IEEE, 2012, pp. 5026–5033. ISBN: 978-1-4673-1737-5. URL: <http://dblp.uni-trier.de/db/conf/iros/iros2012.html#TodorovET12>.